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Experiment-28

Question

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive numbers, negative numbers, and zero elements in an array.
2. Search for an element in an array using **linear search**.
3. Merge two arrays.
4. Exit.

Algorithm

1. Start.
2. Display the menu:
 - Count even, odd, positive, negative and zero elements.
 - Linear search.
 - Merge two arrays.
 - Exit.
3. Input the user's choice.
4. Use a `switch` statement to perform the selected operation.
5. For **Choice 1**:
 - Input the size `n` and elements of the array.
 - Initialize `even`, `odd`, `positive`, `negative`, and `zero` to 0.
 - Traverse the array.
 - If an element is even, increment `even`; otherwise increment `odd`.
 - If the element is positive, increment `positive`.
 - If negative, increment `negative`.
 - Otherwise increment `zero`.
 - Display all counts.
6. For **Choice 2**:
 - Input the size and elements of the array.
 - Input the element to search.
 - Traverse the array from beginning to end.
 - If the element is found, display its position.
 - Otherwise, display "**Element not found**".
7. For **Choice 3**:
 - Input the sizes and elements of two arrays.
 - Copy the elements of both arrays into a third array.

- Display the merged array.
- 8. For **Choice 4**, terminate the program.
- 9. For any other choice, display "**Invalid choice**".
- 10. Repeat the menu until Choice 4 is selected.
- 11. Stop.

Test Case Table

Test Case	Choice	Input	Expected Output
1	1	5 → 2 -3 0 4 -6	Even = 4, Odd = 1, Positive = 2, Negative = 2, Zero = 1
2	2	5 → 10 20 30 40 50, Search = 30	Element found at position 3
3	2	4 → 1 2 3 4, Search = 8	Element not found
4	4	—	Program exits

C Program

```
#include <stdio.h>
```

```
int main()
{
    int choice;
    int n, n1, n2;
    int i, key, found;

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Linear Search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
```

```

case 1:
{
    int even = 0, odd = 0;
    int positive = 0, negative = 0, zero = 0;

    printf("Enter size of array: ");
    scanf("%d", &n);

    int a[n];

    printf("Enter %d elements: ", n);

    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++)
    {
        if(a[i] % 2 == 0)
            even++;
        else
            odd++;

        if(a[i] > 0)
            positive++;
        else if(a[i] < 0)
            negative++;
        else
            zero++;
    }

    printf("Even numbers = %d\n", even);
    printf("Odd numbers = %d\n", odd);
    printf("Positive numbers = %d\n", positive);
    printf("Negative numbers = %d\n", negative);
    printf("Zero elements = %d\n", zero);

    break;
}

```

```

case 2:
{
    printf("Enter size of array: ");
    scanf("%d", &n);

    int a[n];

    printf("Enter %d elements: ", n);

```

```

for(i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}

printf("Enter element to search: ");
scanf("%d", &key);

found = 0;

for(i = 0; i < n; i++)
{
    if(a[i] == key)
    {
        printf("Element found at position %d\n", i + 1);
        found = 1;
        break;
    }
}

if(found == 0)
{
    printf("Element not found\n");
}

break;
}

case 3:
{
    printf("Enter size of first array: ");
    scanf("%d", &n1);

    int a[n1];

    printf("Enter elements of first array: ");

    for(i = 0; i < n1; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter size of second array: ");
    scanf("%d", &n2);

    int b[n2];
    int c[n1 + n2];

    printf("Enter elements of second array: ");

```

```

        for(i = 0; i < n2; i++)
        {
            scanf("%d", &b[i]);
        }

        for(i = 0; i < n1; i++)
        {
            c[i] = a[i];
        }

        for(i = 0; i < n2; i++)
        {
            c[n1 + i] = b[i];
        }

        printf("Merged array: ");

        for(i = 0; i < n1 + n2; i++)
        {
            printf("%d ", c[i]);
        }

        printf("\n");

        break;
    }

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice\n");
    }

} while(choice != 4);

return 0;
}

```

Compilation And Execution

```
(kali㉿kali)-[~]  
$ vi hdoc29.c  
  
(kali㉿kali)-[~]  
$ gcc hdoc29.c -o hdoc29  
  
(kali㉿kali)-[~]  
$ ./hdoc29
```

```
— MENU —  
1. Count even, odd, positive, negative and zero elements  
2. Linear Search  
3. Merge two arrays  
4. Exit  
Enter your choice: 1  
Enter size of array: 5  
Enter 5 elements: 2 -3 0 4 -6  
Even numbers = 4  
Odd numbers = 1  
Positive numbers = 2  
Negative numbers = 2  
Zero elements = 1
```

```
— MENU —  
1. Count even, odd, positive, negative and zero elements  
2. Linear Search  
3. Merge two arrays  
4. Exit  
Enter your choice: 2
```

```
Enter size of array: 5  
Enter 5 elements: 10 20 30 40 50  
Enter element to search: 30  
Element found at position 3
```

```
— MENU —  
1. Count even, odd, positive, negative and zero elements  
2. Linear Search  
3. Merge two arrays  
4. Exit  
Enter your choice: 2  
Enter size of array: 4  
Enter 4 elements: 1 2 3 4  
Enter element to search: 8  
Element not found
```

```
— MENU —  
1. Count even, odd, positive, negative and zero elements  
2. Linear Search  
3. Merge two arrays  
4. Exit  
Enter your choice: 3  
Enter size of first array: 1 2 3  
Enter elements of first array: Enter size of second array: Enter elements of second array: 4 5  
4  
Merged array: 2 4 5 4
```

```
— MENU —  
1. Count even, odd, positive, negative and zero elements
```

```
2. Linear Search  
3. Merge two arrays  
4. Exit  
Enter your choice: 4  
Exiting program...
```